ProgressPulse - Fitness Tracking Application

Made by Jaiveer Bassi for CS360: Database Management Systems

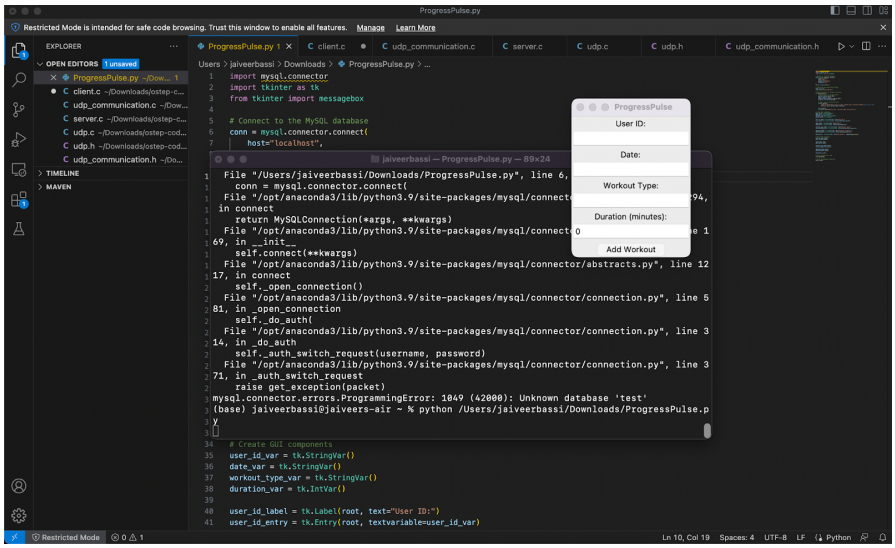
Everything(project and report) is also uploaded to Github: https://github.com/imjbassi/ProgressPulse.
Track your fitness journey with ProgressPulse, a GUI-based fitness-tracking application built using Python and MySQL.

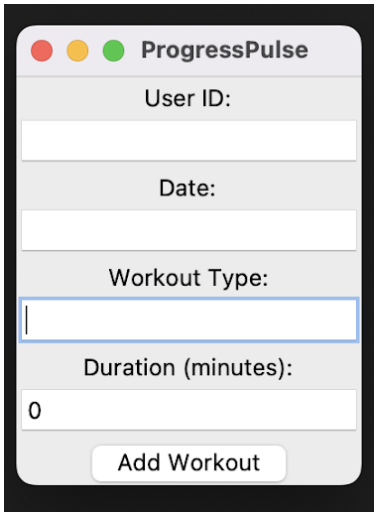
Introduction

ProgressPulse is a fitness-tracking application that allows users to log their workouts, nutrition intake, goals, and more. The application is built with a GUI using Tkinter and stores data in a MySQL database.

Features

- Log and manage workouts with details like date, workout type, and duration.
- Keep track of nutrition intake with meal details and calorie counts.
- Set fitness goals with target values and dates for various types of achievements.
- User-friendly GUI interface for easy data entry and visualization.





Installation

1. Clone the repository: git clone https://github.com/imjbassi/ProgressPulse.git
2. Install required dependencies/libraries: All database elements were used using MySQL: https://www.mysql.com/downloads/
    a. Libraries: pip install -r requirements.txt
    b. To get MySQL to work on Mac, which is what I used, you need Homebrew and need to use brew install mysql-connector-c, pip install mysql-python, and pip install mysql-connector-python
3. Update database configuration (connect function) with your MySQL info.

Usage

1. Run the application: python progresspulse.py
2. Input user data and use the GUI interface to log workouts, nutrition, and goals.
3. Make sure you have a MySQL server running and properly configured.

Code

ProgressPulse.py:

```python
import mysql.connector
import tkinter as tk
from tkinter import messagebox

# MySQL database
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="password",
    database="progresspulsedb"
)
cursor = conn.cursor()

root = tk.Tk()
root.title("ProgressPulse")

def add_workout():
    # Fetch data from the GUI fields
    user_id = user_id_var.get()
    date = date_var.get()
    workout_type = workout_type_var.get()
    duration = duration_var.get()

    # Insert data into Workouts table
    cursor.execute(
        "INSERT INTO Workouts (user_id, date, workout_type, duration) VALUES (%s, %s, %s,
%s)",   (user_id, date, workout_type, duration)
    )
    conn.commit()
    messagebox.showinfo("Success", "Workout added successfully!")

# GUI
user_id_var = tk.StringVar()
date_var = tk.StringVar()
workout_type_var = tk.StringVar()
duration_var = tk.IntVar()

user_id_label = tk.Label(root, text="User ID:")
user_id_entry = tk.Entry(root, textvariable=user_id_var)

date_label = tk.Label(root, text="Date:")
date_entry = tk.Entry(root, textvariable=date_var)

workout_type_label = tk.Label(root, text="Workout Type:")
workout_type_entry = tk.Entry(root, textvariable=workout_type_var)

duration_label = tk.Label(root, text="Duration (minutes):")
duration_entry = tk.Entry(root, textvariable=duration_var)

add_button = tk.Button(root, text="Add Workout", command=add_workout)

# Layout GUI
user_id_label.pack()
user_id_entry.pack()

date_label.pack()
date_entry.pack()

workout_type_label.pack()
workout_type_entry.pack()

duration_label.pack()
duration_entry.pack()

add_button.pack()

# Start GUI event loop
root.mainloop()

# Close db connection
conn.close()
```

Database.sql

```sql
CREATE DATABASE ProgressPulseDB;

USE ProgressPulseDB;

CREATE TABLE Users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(255) NOT NULL,
    email VARCHAR(255) NOT NULL,
    password VARCHAR(255) NOT NULL
);

CREATE TABLE Workouts (
    workout_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    date DATE,
    workout_type VARCHAR(255),
    duration INT,
    FOREIGN KEY (user_id) REFERENCES
Users(user_id)

CREATE TABLE Nutrition (
    nutrition_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    date DATE,
    meal_type VARCHAR(255),
    calories INT,
    FOREIGN KEY (user_id) REFERENCES
Users(user_id)

CREATE TABLE Goals (
    goal_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    goal_type VARCHAR(255),
    target_value INT,
    target_date DATE,
    FOREIGN KEY (user_id) REFERENCES
Users(user_id)
```

Python Code

1. Importing Libraries: The script imports mysql.connector for MySQL database interaction and tkinter for GUI creation.
2. Connecting to Database: The code establishes a MySQL connection and cursor for executing queries.
3. GUI Window Creation: A main window is created with the title “ProgressPulse”.
4. Database Functions: The add_workout() function inserts workout data into the Workouts table.
5. GUI Components: Labels, entry fields, and a button are created for user data input.
6. GUI Layout: Components are arranged vertically for user-friendly input.
7. Event Loop: The GUI event loop is initiated with root.mainloop().
8. Database Connection Close: The MySQL connection is closed after GUI interaction.

Database Schema

- Users (user_id, username, email, password)
- Workouts (workout_id, user_id, date, workout_type, duration)
- Nutrition (nutrition_id, user_id, date, meal_type, calories)
- Goals (goal_id, user_id, goal_type, target_value, target_date)

Disclaimer

ProgressPulse is a personal project created for a class of mine. I doubt I’ll ever update this again.